

Omitted Variable Bias (OVV) and Causal Inference

ECON 490

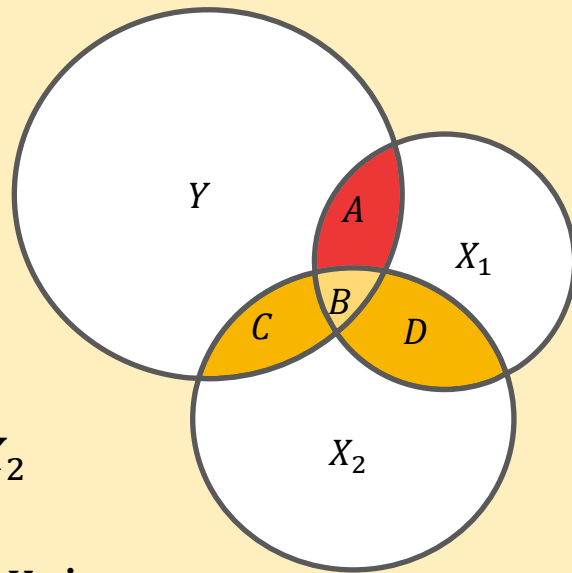
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Slides Overview

In these slides, we'll discuss:

- Defining omitted variable bias (OVV)
- Introducing causal inference
- Fixed effects and OVB

Isolating Variation



A represents covariance of Y and X_1 that is unrelated to X_2

In a regression of Y on X_1 and X_2 , our estimated effect of X_1 is determined by A , the covariance of X_1 and Y that's **unrelated** to X_2

The question for this week – what happens if we **don't** control for X_2 ?

Omitted Variable Bias (OVB)

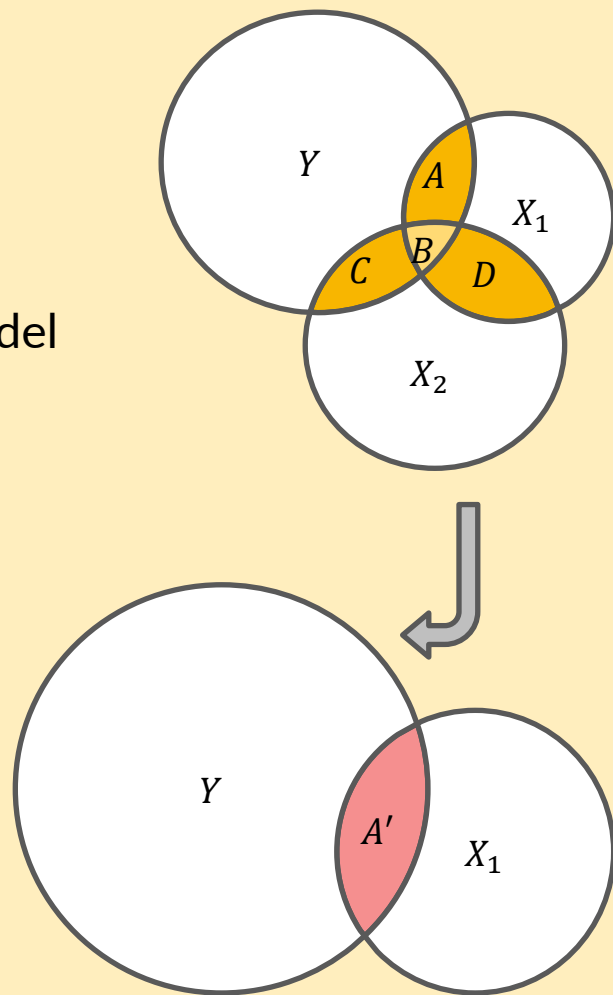
Let's suppose our initial Venn diagram is the “true” model

- What happens if we don't have data for X_2 ?
- Then we're stuck with just observing X_1

The “effect” of X_1 in our **new** diagram is A'

- But that's wrong – it includes B !

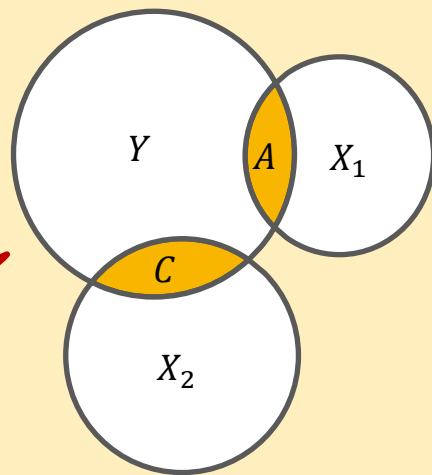
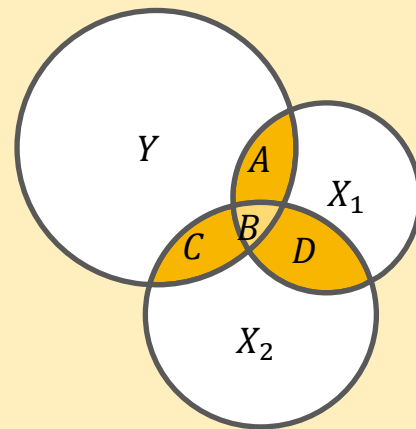
We **incorrectly** estimate the effect of X_1 because we **couldn't** observe X_2 – this is OVB



Two Key Components of OVB

OVB requires **two** conditions:

- 1) Omitted X_2 must covary with Y
- 2) Omitted X_2 must covary with X_1



*If this is the true model,
there's no OVB!*

Direction of Bias

The **bias** in OVB means that our estimated OLS coefficient is wrong

- In other words, our estimated regression coefficient is too big or too small
- The direction of the bias depends on the underlying relationships

Suppose the true model is given by $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + u$

We can refer to the following as the OVB model: $Y = \alpha_0 + \alpha_1^{OVB} X_1 + e$

Direction of Bias Table

Given the definitions from the last slide, we can compare β_1 to α_1^{OVB}

- Table tells us, “What happens if we omit X_2 from our regression?”

	X_1 and X_2 are positively correlated	X_1 and X_2 are negatively correlated
X_2 has a positive effect on Y ($\beta_2 > 0$)	Positive Bias ($\alpha_1^{OVB} > \beta_1$)	Negative Bias ($\alpha_1^{OVB} < \beta_1$)
X_2 has a negative effect on Y ($\beta_2 < 0$)	Negative Bias ($\alpha_1^{OVB} < \beta_1$)	Positive Bias ($\alpha_1^{OVB} > \beta_1$)

NOTE: $OVB = \alpha_1^{OVB} - \beta_1 = \beta_2 \times \delta_1$, where δ_1 is the slope from regressing X_2 on X_1

OVV and Descriptive vs. Causal Statements

Descriptive statement = “A 1-unit increase in X_1 is associated with an $\hat{\alpha}_1$ change in Y ”

- OVB provides helpful context – we know other factors probably matter!
- But claims are still informative – exploring data, relationships, patterns, etc.

Causal statement = “A one-unit increase in X_1 **causes** a β_1 change in Y ”

- Now OVB is a big deal – **can't** make (credible) causal claims if we have OVB
- Causal inference is broadly a set of tools for dealing with OVB

Causal Inference without Randomization

Randomization solves the OVB problem by *randomly* assigning values of X_1

- This ensures X_1 is unrelated to any omitted variables by construction
- Implemented via Randomized Controlled Trials (RCTs)
 - Classic example = pharmaceutical trials

Economics often involves questions where randomization isn't feasible

Causal inference with observational data is the process of “finding” randomization

Causal Inference with Observational Data

“Finding” randomization broadly means one of two things:

1. Identifying situations where treatment assignment is effectively random
2. Controlling for omitted variables to address OVB problem

We can broadly group causal inference models into the above categories

- (1) Includes instrumental variables (IV) & regression discontinuity (RD)
- (2) Includes fixed effects, event studies, diff-in-diff (DiD), etc.

Controlling for Omitted Variables

As a starting point, consider the following:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + u$$

Suppose we *knew* that this was the “true” regression equation

- Would we need to worry about OVB?
- No! Regression identifies *unique* effect of X_1 *controlling* for X_2

Controlling for Omitted Variables (Cont.)

In practice, we almost never know the “true” model

- How do we know what to control for?
- Think about the potential “structure” of omitted variables

Do we think omitted variables...

- Differ across groups in a fixed or constant fashion? (*fixed effects*)
- Differ by group and across time? (*DiD, event studies, etc.*)

We can see how this works by considering an example using fixed effects

Fixed Effects Example

What's the relationship between hours spent studying and test scores?

- Suppose we've got data for two students, A and B, for two tests
- Gives us (1) time studying and (2) test scores

Student	Test	Hours Studying	Test Score
A	1	2	90
A	2	3	96
B	1	5	78
B	2	7	83

Fixed Effects Example (Cont.)

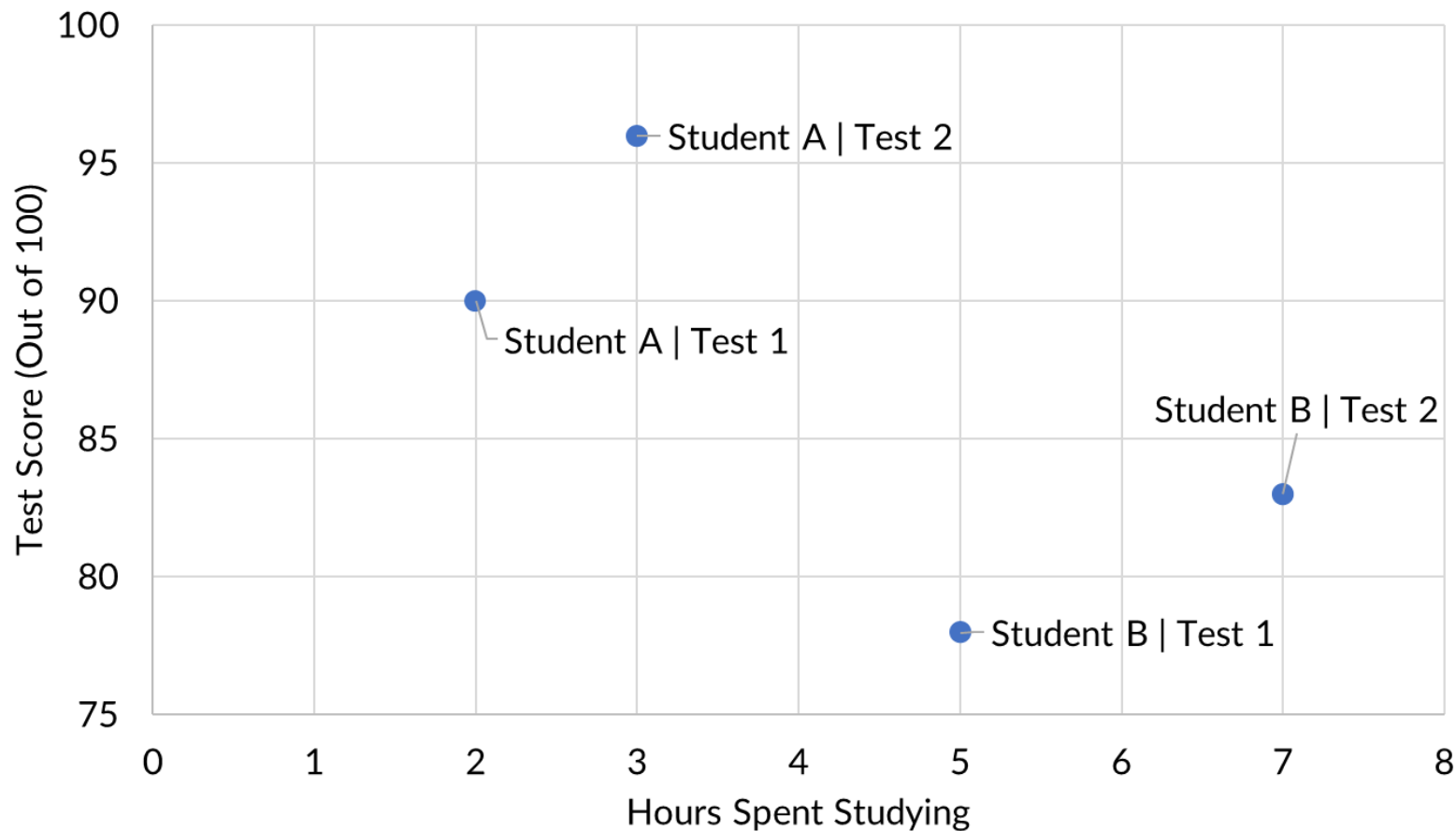
Given our data set, we can run the following regression:

$$Test\ Score_i = \beta_0 + \beta_1 Hours\ Studying_i + u_i$$

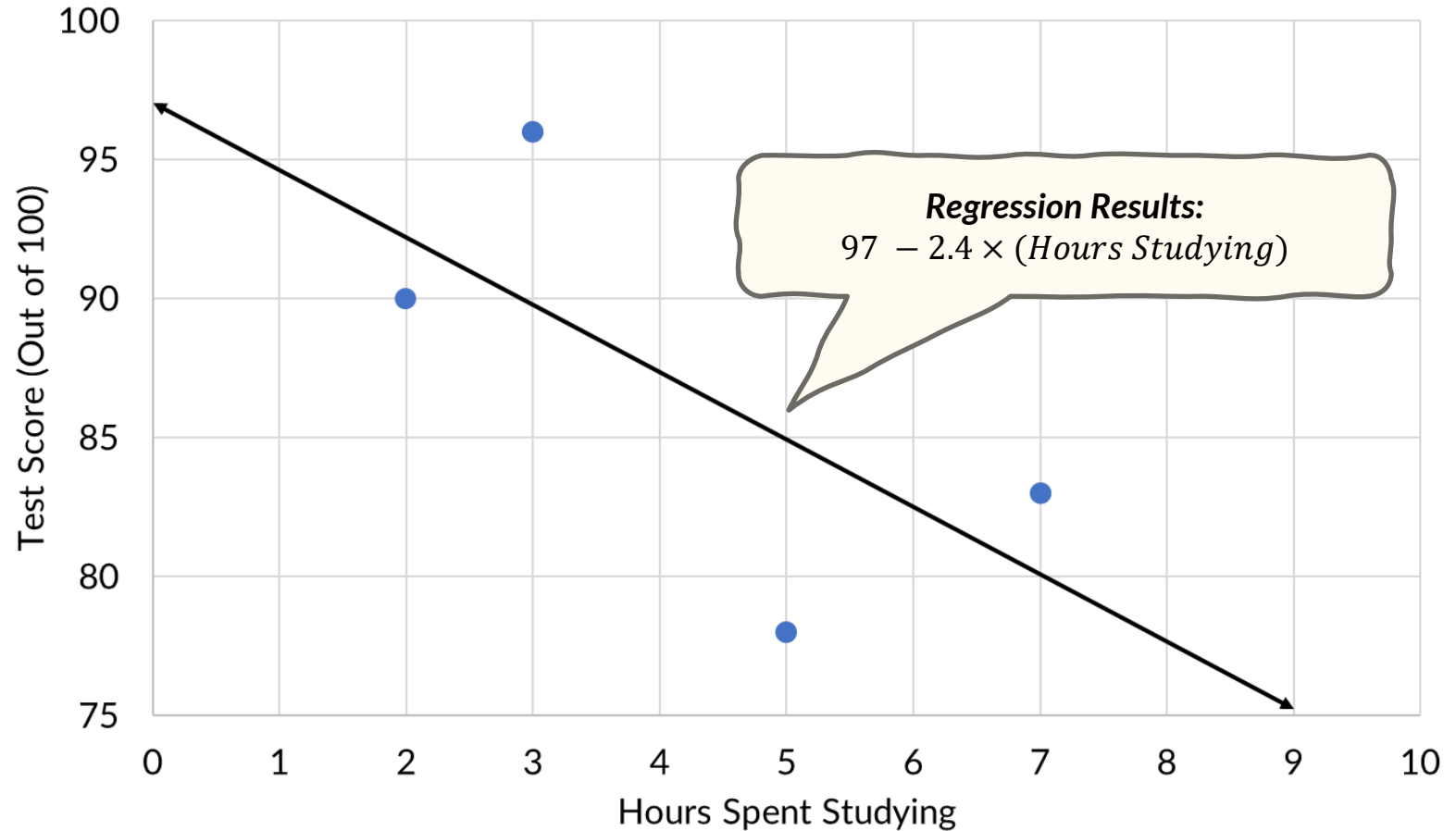
Two ways of interpreting our regression results:

- β_1 tells us the change in avg. scores associated with studying 1 more hour
- We can **predict** test scores given X hours studying using $\hat{\beta}_0 + \hat{\beta}_1 X$

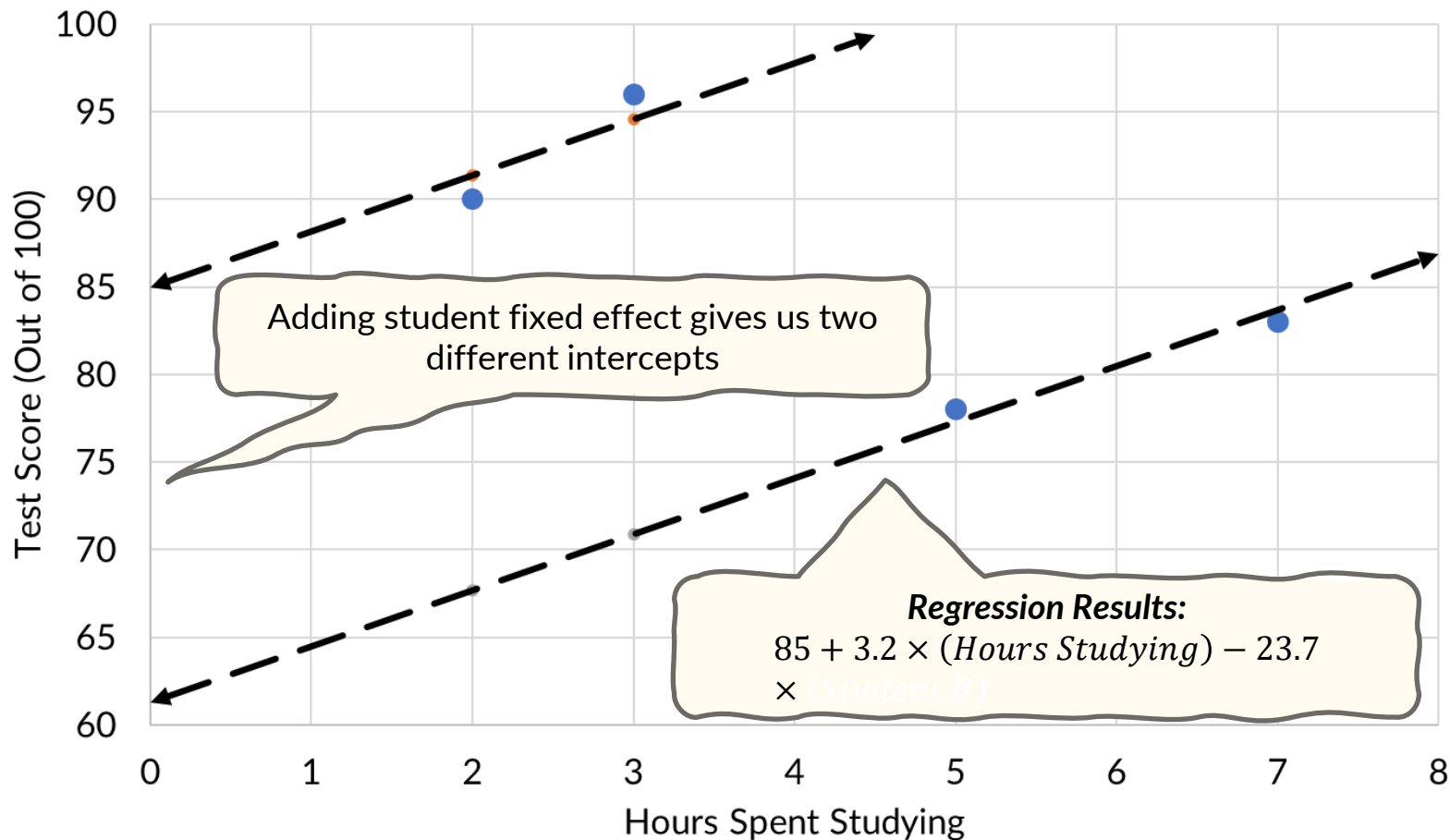
Test Scores as a Function of Hours Studying



Test Scores as a Function of Hours Studying



Test Scores as a Function of Hours Studying



Interpreting Fixed Effects (FEs)

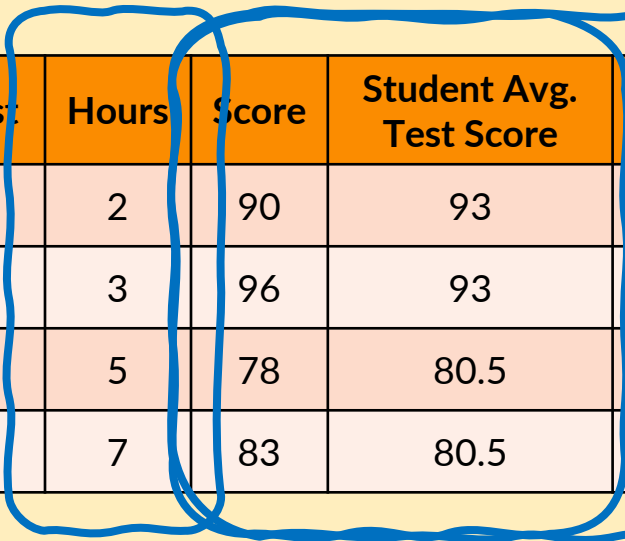
Student FE controlled for avg. differences between students

- In general, we say FEs ***absorb*** variation
- Here, our FE absorbed variation across students in avg. hours + scores

A key question if we're absorbing variation – what variation is left over?

- Refer to this as ***residual variation***
- We can see what this looks like in our example

Residual Variation



Student	Test	Hours	Score	Student Avg. Test Score	Residual Score Variation	Student Avg. Hours	Residual Hours Variation
A	1	2	90	93	-3	2.5	-0.5
A	2	3	96	93	3	2.5	0.5
B	1	5	78	80.5	-2.5	6	-1
B	2	7	83	80.5	2.5	6	1

Bias in Regression

In our example, students systematically differed in both:

- (1) avg. time spent studying and (2) avg. test performance
- This means “student” (broadly defined) was an omitted variable (OV)

Structure of OV in this example = fixed or constant difference in average Y and X

By creating a fixed effect for student, we solved the OVB problem

- Let's us *identify* the causal effect of time studying on test performance

FEs and Causal Inference

FEs are the “building blocks” of research designs like DiD, event studies, etc.

In our example, student FE let us identify the causal effect of studying...

- So long as the **only** source of OVB was avg. differences between students
- This kind of reasoning underlies interpreting all group (2) methods

What might cause OVB? Do we have FEs that absorb that kind of variable?

- We can have fixed effects for time, place, people, etc.
- Adding them can solve OVB... but we still need residual variation!